Major Project Possible Questions

1. The most important questions

These are the questions I would expect an interviewer to ask first:

Project understanding

1. Explain Drishya Vaani in 1–2 minutes.
2. What problem does Drishya Vaani solve?
3. Why did you choose Indian Sign Language (ISL)?
4. What is the complete architecture of your project?
5. Explain the complete flow from camera input to predicted gesture.
6. What was your exact contribution to the project?
7. What challenges did you face while building it?
8. Why did you choose this particular technology stack?
9. What makes your project different from a simple image classification model?
10. What are the limitations of your current system?

---

2. Computer Vision questions

Your resume says you used **MediaPipe Hands** and **OpenCV**, so expect questions here.

MediaPipe

11. What is MediaPipe?
12. Why did you use MediaPipe Hands?
13. How does MediaPipe Hands detect a hand?
14. What are hand landmarks?
15. How many landmarks does MediaPipe Hands generate?
16. What information does each landmark contain?
17. Why are there **21 landmarks**?
18. How did you get **252 dimensions** from the hand landmarks?
19. Why didn't you directly use the raw image as input?
20. What are the advantages of landmark-based features?
21. What happens if the hand is partially occluded?
22. What happens if two hands appear in the frame?
23. How does your system handle different hand orientations?
24. How does lighting affect your system?
25. How does background affect your system?

Very likely follow-up:

"You said 252-dimensional feature vector. Explain exactly how you got 252."

You need to be able to explain this **precisely**, because it is a specific claim on your resume.

---

3. Machine Learning questions

Your resume says:

PCA → StandardScaler → SVM, with 95%+ validation accuracy.

This opens a huge line of questioning.

SVM

26. What is SVM?
27. Why did you choose SVM?

28. How does SVM work?
29. What is a hyperplane?
30. What are support vectors?
31. What is the margin?
32. What is a kernel?
33. What is the kernel trick?
34. Which kernel did you use?
35. Why did you choose that kernel?
36. What is the difference between linear SVM and RBF SVM?
37. What are C and gamma in SVM?
38. What happens if C is too high?
39. What happens if gamma is too high?
40. How does SVM perform with your dataset size?
41. Why didn't you use Random Forest?
42. Why didn't you use XGBoost?
43. Why didn't you use a neural network?
44. Why didn't you use CNN?
45. Why didn't you use a Transformer?

PCA

46. What is PCA?
47. Why did you use PCA?
48. How does PCA work?
49. What is a principal component?
50. What does PCA actually optimize?
51. Why is PCA useful before SVM?
52. Can PCA cause information loss?
53. How did you decide the number of components?
54. Should PCA be performed before or after scaling?

StandardScaler

55. What is standardization?
56. Why do you need StandardScaler?
57. What is the formula for standardization?
58. Why is scaling important for SVM?
59. What happens if you don't scale your features?
60. What is the difference between normalization and standardization?

---

4. Your ML Pipeline

This is **extremely important**.

An interviewer may ask:

"Walk me through your entire ML pipeline."

You should be able to draw:

Camera → OpenCV → MediaPipe → Hand Landmarks → Feature Extraction → Scaling → PCA → SVM → Gesture Prediction

Then explain every stage.

They can also ask:

61. How did you collect the dataset?
62. How many classes did you have?
63. How many samples per class?
64. How did you label the data?
65. How did you split training and validation data?
66. Did you use train-test split?
67. Did you use cross-validation?
68. How did you prevent data leakage?
69. What evaluation metrics did you use?
70. Why is accuracy not always sufficient?
71. What is precision?
72. What is recall?
73. What is F1-score?
74. What is a confusion matrix?
75. Which gestures were hardest to classify?
76. What caused misclassification?
77. How did you handle class imbalance?

---

5. "95%+ accuracy" questions

Because you've put **95%+ validation accuracy** on your resume, expect:

"How did you achieve 95% accuracy?"

Then:

78. What dataset did you use?
79. How large was your dataset?
80. Was the dataset self-created?
81. How did you validate the accuracy?
82. Is it training accuracy or validation accuracy?
83. What was your test accuracy?
84. How did you ensure the model wasn't overfitting?
85. What was your baseline accuracy?
86. What would happen with a completely new person's hand?
87. Does the model generalize to different users?
88. What happens if the camera quality changes?
89. What happens under poor lighting?
90. What happens when the hand moves quickly?

Be particularly careful here: interviewers often challenge an accuracy number to determine whether the candidate actually understands how it was obtained.

---

6. OpenCV questions

Your resume explicitly mentions a **4 FPS real-time frame analysis pipeline using OpenCV decoding**.

Expect:

91. What is OpenCV?
92. Why did you use OpenCV?
93. How do you capture frames?
94. What is a video frame?

95. How do you process a video stream?
96. Why did you choose **4 FPS**?
97. Why not process every frame?
98. How does reducing FPS improve performance?
99. What is frame skipping?
100. How do you reduce latency?
101. What happens if ML inference is slower than incoming frames?
102. How do you handle dropped frames?
103. What is the difference between FPS and latency?

---

7. WebRTC questions — VERY IMPORTANT

This is probably the most difficult networking portion of your project.

Your resume says:

WebSocket signaling + SDP/ICE + STUN/TURN + P2P video calls + symmetric NAT.

Be ready for:

104. **What is WebRTC?**
105. Why did you use WebRTC?
106. What is peer-to-peer communication?
107. How does WebRTC establish a connection?
108. What is signaling?
109. Why is signaling required?
110. What is SDP?
111. What information does SDP contain?
112. What is an SDP offer?
113. What is an SDP answer?
114. What is ICE?
115. What is an ICE candidate?
116. What is STUN?
117. Why do we need STUN?
118. What is TURN?
119. Why do we need TURN if STUN exists?
120. What is NAT?
121. What is symmetric NAT?
122. Why is symmetric NAT problematic for P2P communication?
123. How does TURN solve the problem?
124. What happens if STUN fails?
125. What happens if both peers are behind restrictive NAT?
126. What is ICE candidate gathering?
127. What is ICE negotiation?

---

8. WebSocket questions

You specifically mention a **dual-channel FastAPI WebSocket signaling server**.

Expect:

128. What is WebSocket?
129. Why WebSocket instead of HTTP?

130. Difference between HTTP and WebSocket?
131. Is WebSocket full-duplex?
132. How does a WebSocket connection work?
133. What is the WebSocket handshake?
134. Why is WebSocket useful for signaling?
135. What information do you send through WebSocket?
136. What are your two WebSocket channels?
137. How do you identify different clients?
138. How do you handle connection/disconnection?
139. What happens if a WebSocket connection drops?
140. How do you handle multiple users simultaneously?

---

9. FastAPI questions

Your project uses FastAPI for serving the ML system and WebSocket communication.

Prepare:

141. What is FastAPI?
142. Why FastAPI instead of Flask?
143. What is ASGI?
144. What is Uvicorn?
145. Difference between WSGI and ASGI?
146. What does async mean?
147. What is asynchronous programming?
148. Why is async useful for your project?
149. Can CPU-heavy ML inference actually benefit from async?
150. How does FastAPI handle concurrent requests?
151. What are Pydantic models?
152. How did you validate incoming requests?
153. How did you handle errors?
154. How did you structure your FastAPI application?

---

10. Security questions

You have explicitly written:

API-key authentication, CORS, and payload guards.

Therefore interviewers can absolutely ask:

155. What is authentication?
156. What is authorization?
157. How does API-key authentication work?
158. Why did you use API keys?
159. Where did you store the API key?
160. Should an API key be hardcoded?
161. What is CORS?
162. Why was CORS required in your project?
163. What is a payload?
164. What is a payload guard?
165. Why do you need payload validation?

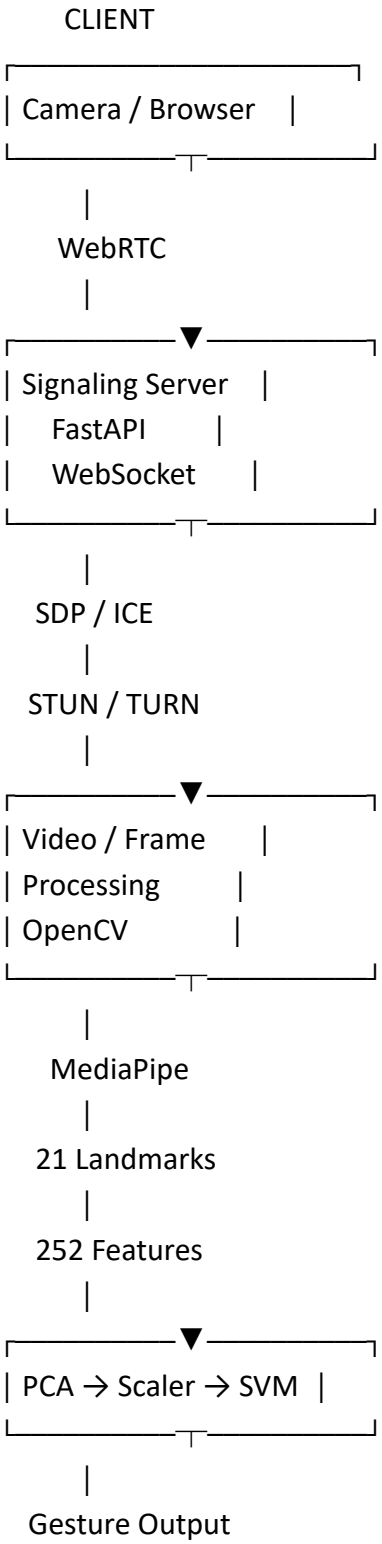
166. What happens if a malicious user sends a huge payload?
167. How can you prevent denial-of-service through oversized requests?
168. What other security measures would you add?
169. Difference between authentication and authorization?
170. What is HTTPS and why is it important?

---

11. Architecture questions

You should be able to draw your architecture on a whiteboard.

Something like:



Then they can ask:

171. Why did you separate signaling and inference?

172. Why WebRTC instead of sending video through HTTP?
173. Why WebSocket instead of REST for signaling?
174. Where does the ML model run?
175. Is your architecture scalable?
176. What is the bottleneck?
177. What happens when 100 users connect?
178. How would you deploy this system?
179. How would you scale the inference server?
180. Would you use GPUs?
181. How would you load-balance inference requests?

---

12. Real-time system questions

This is another area where your project becomes interesting.

182. What does "real-time" mean in your project?
183. What is latency?
184. What is throughput?
185. What is FPS?
186. How did you reduce latency?
187. Why 4 FPS?
188. What is the ML inference time?
189. What happens when network latency increases?
190. What happens when packet loss occurs?
191. How would you improve real-time performance?
192. How would you measure end-to-end latency?

---

13. Questions comparing technologies

Interviewers love these.

WebRTC vs WebSocket

193. Why WebRTC instead of WebSocket for video?

REST vs WebSocket

194. Why not use REST API for real-time communication?

SVM vs CNN

195. Why SVM instead of CNN?

MediaPipe vs OpenCV

196. What is the difference between MediaPipe and OpenCV?
197. Why do you need both?

STUN vs TURN

198. What is the difference between STUN and TURN?

FastAPI vs Flask

199. Why FastAPI?

PCA vs no PCA

200. What happens if you remove PCA?

---

14. Deep-dive questions they may ask if the interviewer is strong

These are the questions I'd consider **advanced-level preparation**:

201. Explain the mathematical intuition behind SVM.
202. Explain how PCA calculates principal components.
203. What is covariance matrix?
204. What are eigenvectors and eigenvalues?
205. Why are eigenvectors relevant to PCA?
206. Why does scaling affect PCA?
207. Explain the bias-variance tradeoff.
208. What is overfitting?
209. How can you detect overfitting?
210. What is cross-validation?
211. Explain the difference between validation and test datasets.
212. How would you handle temporal correlation in video frames?
213. Why can randomly splitting frames cause data leakage?
214. How would you design a user-independent train/test split?
215. How would you make the system robust to different skin tones, lighting conditions and camera angles?
216. How would you recognize continuous sign-language sentences rather than isolated gestures?
217. How would you incorporate temporal information?
218. Would you use LSTM/GRU/Transformer for continuous signing?
219. How would you distinguish two gestures that have similar static hand poses?
220. How would you handle left-hand vs right-hand signing?

15. Questions about the project that you MUST be able to answer

If this is for a placement/evaluation, I would make these your **top 20**:

Priority	Question
1	Explain Drishya Vaani
2	Explain complete architecture
3	Explain end-to-end data flow
4	Why MediaPipe?
5	Why 21 landmarks?
6	How did you get 252 features?
7	Why SVM?
8	Explain SVM
9	Why PCA?
10	Why StandardScaler?
11	Explain PCA → StandardScaler → SVM
12	How did you achieve 95%+ accuracy?
13	What is WebRTC?
14	What is signaling?

Priority	Question
🔴	15 Explain SDP and ICE
🔴	16 What are STUN and TURN?
🔴	17 Why WebSocket?
🔴	18 Why FastAPI?
🔴	19 What is CORS?
🔴	20 What are payload guards?

One particularly important point

Because your resume contains **very specific technical claims**, don't just memorize definitions. The interviewer can keep drilling down:

"You used SVM."

→ Why SVM?
→ Which kernel?
→ Why that kernel?
→ What is C?
→ What is gamma?
→ How did you tune them?
→ How did you validate the model?

Or:

"You used WebRTC."

→ Why WebRTC?
→ How does it establish connection?
→ What is signaling?
→ What is SDP?
→ What is ICE?
→ What is STUN?
→ Why TURN?
→ What is symmetric NAT?

So **every technology you put on your resume should be prepared at least 3–4 levels deep.**